# GreenTwin

## System design: five parts, their contracts, data and build plan

Megathon X: Singularity · Lloyds Problem Statement 2: Green Home Score and Retrofit Recommendation Engine · Team MegaMind · contracts schema 0.4.0

> **In one line.** A Bayesian-calibrated physics twin of every home, which you can see in 3D, explain in plain language and act on, where every number is backed by physics and the range it comes with.

**Audience:** the build team. This document is the high-level design: the five parts, how they connect, the data that feeds them and the build plan. The companion **LLD.pdf** holds every contract field, interface signature, algorithm, table schema and download step. Binding rules: AGENTS.md and contracts/.

| Section | What it covers |
|---|---|
| 1. Philosophies | The six principles every part follows; how the brief's core rule is enforced |
| 2. The five parts | What each part answers, what is inside it, coupled core vs independent parts |
| 3. Architecture | End-to-end flow, the what-if loop, technology choices |
| 4. Part designs | Purpose, components, models, contracts in/out and done-when for each part |
| 5. Contracts and interfaces | The contract and Protocol each part owns (full detail in LLD.pdf) |
| 6. Data | Datasets and how to get them, the EPC live tables, joins, synthetic data, day-1 plan |
| 7. Uncertainty | Every number as a full distribution |
| 8. Key mechanisms | Improvement ladder, improvement potential |
| 9. Evaluation and submission | Metrics vs baselines, PS coverage, submission items |
| 10. Build plan | Work breakdown, checkpoints, coding guidelines, open decisions |

# 1. Core philosophies

| # | Philosophy | What it means in the system |
|---|---|---|
| 1 | **Physics is the source of truth** | Every energy, savings, carbon and rating number comes from the Level-1 digital twin (Part 1). ML sets priors (EBM), estimates value and checks results; it never produces a customer's savings, so the system handles retrofit combinations no dataset has seen. |
| 2 | **Bayesian in thought** | Each home is calibrated on its own meter readings. Every output is a full distribution (mean, median, mode, std, percentiles, 90% interval, tail probabilities) plus P(Poor), P(suitable), P(on front). Missing data never becomes zero; it widens the range. |
| 3 | **Rerun, never add up** | A retrofit is a change to the twin's parameters; every what-if and package reruns the model. Interactions come from the physics. Savings are always per fuel. |
| 4 | **Grounded explanation** | The LLM only quotes tool outputs; a validator rejects any other number; Jev only routes questions and screens feasibility. No backing → "unavailable". |
| 5 | **Explainability first** | Score waterfall, heat loss per element, EBM contributions, value coefficients, a reason on every recommendation or veto. |
| 6 | **See it to believe it** | A living 3D twin: heat loss glows, retrofits appear, numbers update live, all driven by the same calibrated model. |

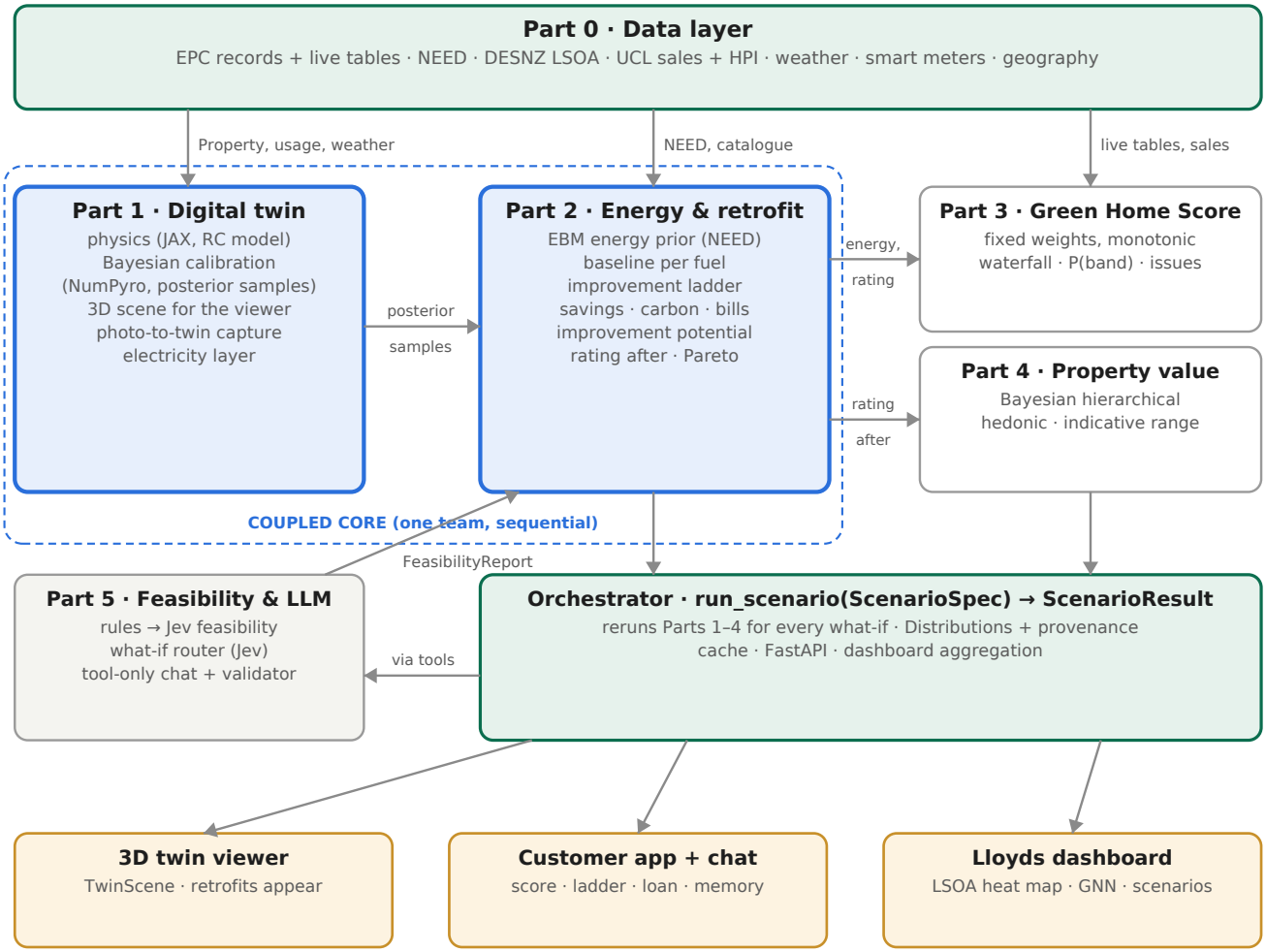
### The brief's core rule, enforced by design

"*The score, savings, carbon and value figures come from documented models and calculations. The LLM only explains them... For a what-if request, the relevant models must first be rerun.*" (PS §1)

- Only Parts 1–4 produce numbers, each with provenance naming its model.
- Every what-if goes through **run_scenario** before the LLM sees anything.
- The LLM's answer passes a numeric validator → regenerate once → template explainer.

## 2. The five parts

GreenTwin is five parts on top of a shared data layer and a thin orchestrator. Parts 1 and 2 form the **coupled core**: every energy number runs through the twin, so they are built by one team in sequence. Parts 3, 4 and 5 (and the data layer and UI) are **independent**: each codes against its interface and a stub, and connects to the core at a few fixed points.



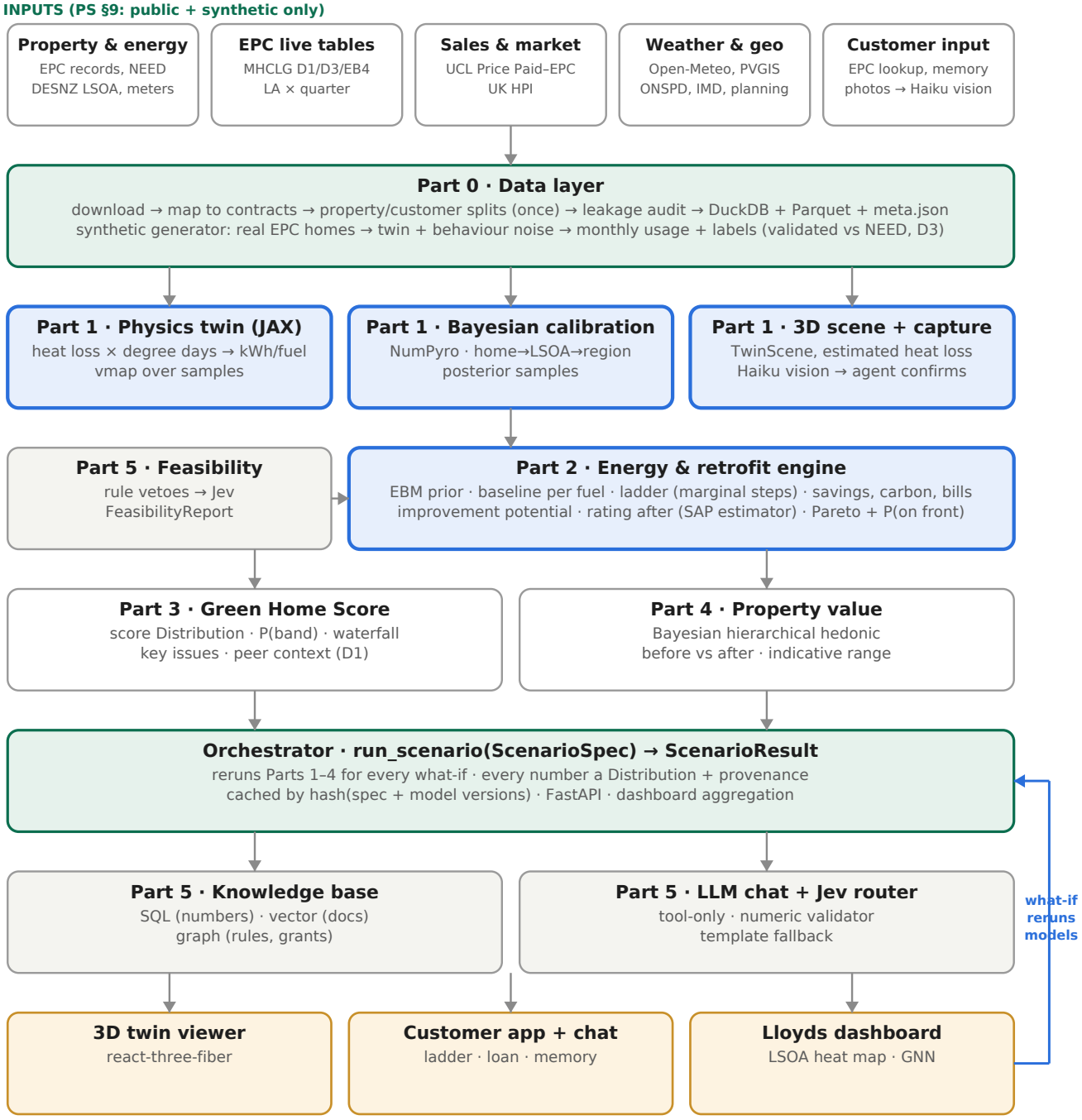

Blue = coupled core (every energy number runs through the twin). Green = data + orchestration. Grey = LLM side (reads only via tools).
Parts 0, 3, 4, 5 and the UI are independent: each codes against its interface (greentwin/interfaces.py) and a stub, then plugs in.

| Part | Answers (PS) | Inside | Coupling |
|---|---|---|---|
| **1 · Digital twin** | the physics behind §3, §5, §6 | JAX physics twin · NumPyro Bayesian calibration · 3D scene · photo-to-twin capture · electricity layer | Core |
| **2 · Energy & retrofit** | §3 potential, §4 retrofits, §5 savings, §6 carbon | EBM energy prior · baseline · improvement ladder · savings, carbon, bills · improvement potential · rating after · Pareto | Core |
| **3 · Green Home Score** | §2 score, §3 poor homes | fixed-weight score · sub-scores · waterfall · key issues · projected score · peer context | Independent |
| **4 · Property value** | §7 value | Bayesian hierarchical hedonic model · before vs after · green twins | Independent |
| **5 · Feasibility & LLM** | §1 core rule, §4 feasibility, §8 what-if, §10 memory | rules → Jev feasibility · what-if router · tool-only chat + validator · knowledge base · memory | Independent |

### The four connection points

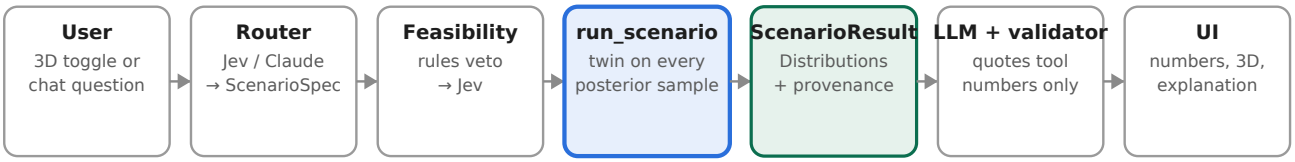
| From → to | Contract | Until it exists, use |
|---|---|---|
| Part 2 EBM → Part 1 calibration | EnergyPrior (prior mean per fuel) | RdSAP defaults |
| Part 5 feasibility → Part 2 ladder | FeasibilityReport | rules only |
| Part 2 rating → Parts 3, 4 | RatingEstimate (estimated SAP after) | stub: band + 1 |
| Orchestrator → Part 5 LLM, UI | ScenarioResult, TwinScene | contracts/examples/*_MOCK.json |

# 3. Architecture (end to end)




Every arrow is a contract in contracts/models.py; the experience layer never computes, it renders ScenarioResult / TwinScene.

## The what-if loop




Low router confidence → clarifying question. Validator failure → regenerate once → template explainer. Missing data → "unavailable" + reason.

## Technology choices

| Concern | Choice |
|---|---|
| Data + storage | Python 3.12, pandas/Polars, DuckDB + Parquet with a meta.json per file |
| Physics twin | **JAX** (jax.numpy): runs inside the NumPyro model so NUTS/SVI get exact gradients; jax.vmap over posterior samples. Monthly steady-state RC model; differentiable RC graph (diffrax) later |
| Bayesian | NumPyro: NUTS offline for the hierarchy, SVI per home (< 1 s); ArviZ diagnostics |
| Electricity layer | Half-hourly: smart-meter load-shape archetypes + heat pump + PVGIS PV − battery |
| Learned models | EBM (InterpretML) energy prior, XGBoost baseline; GraphSAGE GNN (PyTorch Geometric) on the LSOA graph; Bayesian hedonic value model (NumPyro) |
| Photo-to-twin | Claude Haiku 5.5 (claude-haiku-5-5) vision + structured outputs; react-konva editor |
| LLM + router | Claude tool use (LLM_MODEL, team default claude-sonnet-5-5) + numeric validator + template fallback; Jev System 1 if a key exists, else Claude structured output |
| API + front end | FastAPI; React + Vite + TypeScript, react-three-fiber + drei, Recharts, MapLibre + deck.gl, Zustand; types generated from the contracts |

# 4. Part designs

## Part 0 · Data layer

PS §9 · feeds every part

**Purpose.** Clean, licensed, split, joinable real data and every external factor traceable to a cited source.

- Download scripts write raw files untouched, then Parquet + meta.json (DatasetMeta) per source.
- EPC record → Property mapper: regex rules turn EPC text (walls, roof, glazing, heating, controls) into contract enums; ≥ 95% mapped per field; unmapped values logged.
- EPC live tables (MHCLG D1/D3/EB4) → AreaEPCStats per local authority × quarter: calibration targets, validation checks, peer context and dashboard layers (section 6).
- Splits: 70/15/15 by customer, else property, seed 42, assigned once; spatial-block splits for area models; leakage audit (epc.potential_* are labels only).
- Synthetic generator (uses the Part 1 twin): real EPC homes + NEED-calibrated behaviour → monthly usage with gaps and conflicts + reference labels; reweighted to EPC live-table band and type mixes; validated vs NEED.

| | |
|---|---|
| **Models (primary · baseline)** | — (data engineering); EPC mapper, k-means load profiles |
| **Consumes** | public datasets, customer input |
| **Produces** | Property, EnergyUsageMonth, SaleRecord, WeatherMonth, AreaEPCStats, LSOAConsumption, LoadProfileArchetype, EPCRecommendation, ReferenceLabel, DatasetMeta |
| **Done when** | No id spans two splits; mapper coverage ≥ 95%; our EPC counts reconcile with live table D1; every table has meta.json; DATA section of the LLD complete. |

## Part 1 · Digital twin (physics, Bayesian, 3D)

PS §3, §5, §6 physics · coupled core

**Purpose.** A documented physics model of each home, calibrated on its own bills, that every energy number runs through, and that the customer can see in 3D.

- Physics (JAX): heat loss Σ U·A + ventilation × degree days → space heat; hot water; baseload; fuel via boiler efficiency or heat-pump SCOP; PV cuts imports. Calibrated and standard (SAP occupancy) modes.
- Bayesian calibration (NumPyro): heat-loss multiplier, setpoint, baseload, efficiency; hierarchy home → LSOA → region; Student-t likelihood on monthly meters; missing months latent; SVI per home, NUTS offline.
- 3D scene: parametric house (or captured rooms) with surfaces coloured by estimated heat-loss share and retrofit slots; never calculates.
- Photo-to-twin: Haiku vision drafts rooms + element tags with confidence; agent confirms in a react-konva editor; room geometry improves areas and the 3D house.
- Electricity layer: half-hourly profile for PV self-consumption, battery and time-of-use bills.

| | |
|---|---|
| **Models (primary · baseline)** | Level-1 RC twin (physics) · Bayesian calibration (vs degree-day OLS, LightGBM quantile) |
| **Consumes** | Property, EnergyUsageMonth, WeatherMonth, EnergyPrior, LoadProfileArchetype |
| **Produces** | TwinParams, CalibrationResult, PosteriorSamples (in-process), ElementHeatLoss, TwinScene, CaptureResult, ElectricityProfileSummary |
| **Done when** | SH1–SH3 within ±15% of metered totals; monotone sweep passes; 90% coverage ≈ 90% on held-out homes; standard-mode kWh/m² tracks EPC live table D3 by local authority. |

## Part 2 · Energy calculation & retrofit efficiency

PS §3, §4, §5, §6 · coupled core

**Purpose.** Turn the calibrated twin into the numbers the brief asks for: baseline energy per fuel, what each retrofit adds, savings, carbon, bills, improvement potential, the estimated rating after, and the best packages.

- Energy prior: EBM on NEED predicts kWh/m² per fuel (prior for homes without meters) with feature contributions.
- Baseline: per-fuel Distributions; intensity = annualised metered energy per m² (the brief's definition).
- Improvement ladder: feasible retrofits in fabric-first order; each step's marginal effect on every posterior sample; P(suitable) = P(feasible) × P(net benefit); probabilistic verdicts.
- Improvement potential: all feasible retrofits in one run; null + reason if the baseline is 0.
- Rating after: SAP 2012 formula on the twin's standard mode, anchored to the recorded SAP.
- Pareto: enumerate feasible packages; P(on front) across samples; knee.

| | |
|---|---|
| **Models (primary · baseline)** | EBM energy prior (vs NEED median, XGBoost) · SAP estimator (vs median band shift) · Pareto (vs greedy kWh/£) |
| **Consumes** | PosteriorSamples, FeasibilityReport, catalogue, emission factors, tariffs |
| **Produces** | EnergyPrior, EnergyBreakdown, RetrofitEffect, ImprovementPotential, RatingEstimate, ParetoPoint |
| **Done when** | SH2 shows electricity rising after a heat pump; package = one run; potential null at baseline 0; rating change validated on EPC before/after pairs; precision@k vs EPC assessor recommendations. |

## Part 3 · Green Home Score

PS §2, §3 · independent

**Purpose.** Answer "how green is my home" with the brief's fixed weights, a transparent waterfall and the issues that matter.

- S = Σ w·s from contracts/score_mapping.json; seven monotonic, documented sub-scores; rounded; five bands.
- Energy efficiency from the recorded SAP (or the estimated rating for a projected score); intensity from annualised metered kWh/m².
- Score Distribution and P(each band) from posterior samples; key issues with evidence; peer context from the EPC live tables (local-authority band shares, D1).

| | |
|---|---|
| **Models (primary · baseline)** | Deterministic scoring (no training) |
| **Consumes** | Property, EnergyBreakdown, RatingEstimate, AreaEPCStats |
| **Produces** | ScoreResult (components, KeyIssue list, band probabilities) |
| **Done when** | Weights sum to 1; monotonicity sweep; every heating type mapped; projected ≥ current for improvement-only packages. |

## Part 4 · Property value

PS §7 · independent

**Purpose.** An indicative value range, before vs after, accounting for location, size, type and market.

- Bayesian hierarchical hedonic regression on real UK sales: log(price ÷ HPI) = pooled location effects (LSOA within LA within region) + size + type + age + tenure + β(region × type) × SAP.
- Same home at its recorded vs estimated-after rating, applied to the current value estimate; full distribution; wording always "indicative"; range including 0 → no clear effect.
- Green twins: matched greener comparable homes shown as ranges.

| | |
|---|---|
| **Models (primary · baseline)** | Bayesian hedonic (vs OLS without the rating) |
| **Consumes** | Property, SaleRecord (training), RatingEstimate |
| **Produces** | ValueImpact, GreenTwin |
| **Done when** | Held-out coverage and pinball loss beat the baseline; β compared with Nationwide / DECC 2013; planted-premium recovery reported as a method check only. |

## Part 5 · Feasibility rules engine, Jev, LLM explanation and chat

PS §1, §4, §8, §10 · independent

**Purpose.** Decide what is possible for a home, route what-if questions, and explain computed numbers without ever inventing one.

- Feasibility: rule vetoes from contracts/feasibility_rules.json always win; Jev classifies survivors with recalibrated confidence; low confidence → needs survey; Claude structured output without a Jev key.
- Router: Jev intent + ScenarioSpec slots; below the threshold → clarifying question.
- Chat: tool-only Claude (get_home_facts, get_results, run_scenario, search_docs, query_graph), numeric validator, regenerate once, template fallback; knowledge base (SQL, vector, graph); customer memory.

| | |
|---|---|
| **Models (primary · baseline)** | Jev threshold calibration (isotonic) · LLM evaluated on ≥ 30 adversarial scenarios |
| **Consumes** | Property, catalogue, CustomerMemory, ScenarioResult (via tools) |
| **Produces** | FeasibilityReport, WhatIfRoute, ChatResponse, ExplanationCheck, CustomerMemory |
| **Done when** | Rule veto beats Jev; 0 unvalidated numbers; groundedness and accuracy ≥ 90%, unsupported claims < 5%; works with no keys. |

# 5. Contracts and interfaces

Parts exchange data only through Pydantic contracts (contracts/models.py, schema 0.4.0; TypeScript mirror generated into contracts/types.ts) and call each other only through the Protocols in greentwin/interfaces.py. Every field and signature is listed in LLD.pdf, generated from the code.

| Part | Interface (Protocol) | Key methods | Owns these contracts |
|---|---|---|---|
| 0 Data | DataLayer | load_property, load_usage, load_weather, load_sales, load_area_epc_stats, load_lsoa_consumption, load_load_profiles, catalogue | Property, EnergyUsageMonth, SaleRecord, WeatherMonth, AreaEPCStats, LSOAConsumption, LoadProfileArchetype, EPCRecommendation, ReferenceLabel, DatasetMeta |
| 1 Twin | DigitalTwin | params_from_property, apply_retrofits, simulate, calibrate, element_heat_loss, electricity_profile, scene, capture | TwinParams, CalibrationResult, ElementHeatLoss, TwinScene, SceneSurface, RetrofitSlot, FloorPlan, Room, ElementTag, CaptureResult |
| 2 Energy | EnergyRetrofitEngine | energy_prior, baseline, ladder, package, improvement_potential, estimate_rating, pareto | EnergyPrior, EnergyBreakdown, MonthlyEnergy, ElectricityProfileSummary, RetrofitEffect, ImprovementPotential, RatingEstimate, ParetoPoint |
| 3 Score | GreenHomeScorer | score | ScoreResult, ScoreComponent, KeyIssue |
| 4 Value | ValueModel | value_impact, green_twins | ValueImpact, GreenTwin |
| 5 Feasibility & LLM | FeasibilityEngine, Router, Explainer | assess, route, chat | Feasibility, FeasibilityReport, WhatIfRoute, ChatRequest, ChatResponse, ExplanationCheck, Citation, CustomerMemory |
| Orchestrator | Orchestrator | run_scenario, dashboard_areas | ScenarioSpec, ScenarioResult, DashboardArea |
| Shared | contracts.summarise | summarise, exact, category_probabilities | Distribution, Provenance, FeatureContribution |

**Rule for every part:** uncertain numbers are Distributions built with contracts.summarise; every result carries Provenance; missing data is None (never 0); savings are per fuel; packages are re-run, never summed. A test fails if types.ts drifts from models.py.

# 6. Data

No public dataset links property features, monthly meter readings and sale prices for the same homes. So: **real data for every part that exists publicly**, joined at the level each dataset allows, plus **physics-generated synthetic data only for the missing piece** (monthly readings per home and the labels), calibrated and validated against real aggregates.

## Datasets and how to get them

| Dataset | Used by | Access | Licence |
|---|---|---|---|
| **EPC domestic certificates + recommendations** | Part 0 homes · Part 3 score inputs · Part 2 rating check (certificate history) · Part 5 rules · recommendation labels | Bulk CSV download + developer API (OpenAPI 3, bearer token) | OGL v3 except where stated (see the service's licensing restrictions page) |
| **EPC live tables (statistics)** | Part 0 calibration targets + download check · Part 1 twin validation · Part 3 peer context · dashboard | Direct ODS download, quarterly (data to June 2026, published 30 July 2026) | OGL v3 |
| **NEED anonymised data 2025** | Part 2 EBM energy prior (training) · Part 0 synthetic behaviour calibration and validation | Direct download: 50,000-row and 4,000,000-row property-level samples + metadata | OGL v3 |
| **Sub-national domestic electricity and gas by LSOA** | GNN area-model labels · dashboard observed layer · synthetic area check | Direct XLSX download (LSOA domestic electricity, LSOA domestic gas) | OGL v3 |
| **UCL linked Price Paid + EPC, 1995–2024 (ReShare 857911)** | Part 4 Bayesian hedonic value model (training) · green twins | Open download, no registration: 857911_bundle_1.zip (3 GB) + bundle_2.zip (4 GB) | CC BY 4.0 |
| **HM Land Registry Price Paid Data** | Part 4 recent sales after Oct 2024 (needs linking to EPC) | Direct CSV (complete file, yearly, monthly) | OGL v3 with attribution (address data has extra rights notes) |
| **UK House Price Index** | Part 4 market adjustment (deflate prices to today) | CSV download | OGL v3 |
| **Smart meter energy consumption, London households (Low Carbon London)** | Part 1 electricity layer: half-hourly load-shape archetypes | Direct download (combined CSV ≈ 10 GB unzipped, ~167M rows; tariff file) | Open (see the London Datastore page) |
| **Open-Meteo Historical Weather API (ERA5)** | Part 0/1 monthly heating degree days and irradiance per postcode district | REST GET, JSON; cache every response | Data CC BY 4.0; cite Open-Meteo and ERA5 with the Copernicus acknowledgement |
| **Met Office HadUK-Grid** | Documented degree days and 30–50-year regional trends (dashboard) | CEDA archive download | OGL |
| **PVGIS 5.3 API** | Part 1 PV yield (monthly PVcalc, hourly seriescalc) | REST GET at https://re.jrc.ec.europa.eu/api/v5_3/{PVcalc\|seriescalc\|tmy} | Free use with attribution to EU JRC PVGIS |

| Dataset | Used by | Access | Licence |
|---|---|---|---|
| **ONS Postcode Directory (ONSPD)** | The join key: postcode → LSOA 2021, local authority, region, lat/lon | CSV download (May 2026 release; August 2026 user guide published) | OGL (contains OS and Royal Mail data; attribution required) |
| **English Indices of Deprivation 2025** | Features for the EBM prior, value model and GNN; dashboard | Direct download (File 1 IMD; download-all tool at deprivation.communities.gov.uk) | OGL v3 |
| **LSOA 2021 boundaries** | Dashboard maps; LSOA adjacency graph for the GNN | GeoJSON / GeoPackage download | OGL |
| **Conservation areas + listed buildings** | Part 5 feasibility rules (EWI, glazing restrictions) | Direct CSV / GeoJSON / Parquet | OGL v3 (attribution: Historic England, Ordnance Survey) |
| **Off-gas-grid indicator** | Part 5 rules and grants (BUS off-gas uplift); dashboard layer | EPC field per home; postcode lists as XLSX/CSV | EPC: OGL; postcode lists: check terms (CSE 2017 listing has no licence) |
| **Electrification of Heat heat-pump performance (SN 9050 / 9210 / 9209)** | Part 1 heat-pump SCOP priors by house type | SN 9050 cleansed data: open, summary dataset downloadable; full cleansed data by FTP order | UK Data Service terms (check per study) |
| **English Housing Survey energy tables** | Part 0 stock distributions to fill gaps in the synthetic population | Direct XLSX tables | OGL v3 |
| **CubiCasa5k floor plans** | Part 1 photo-to-twin evaluation (room polygons, areas) | Zenodo download | Check the repository LICENSE; treat as non-commercial until confirmed |
| **Octopus Energy public products API (time-of-use tariff citation)** | Part 2 time-of-use bill figures (cite one published tariff) | REST GET https://api.octopus.energy/v1/products/ ; product details list tariff unit-rate links | Octopus API terms; cite tariff, URL and access date in tariffs.json |

## Start here (day 1, in this order)

1. Register GOV.UK One Login and get EPC bulk access (the slowest step); use cached samples meanwhile.
2. Download the EPC live tables D1, D3, EB1, EB3, EB4 (ODS, minutes) and build area_epc_stats.
3. Download NEED 2025 (50k now, 4M for training) with its metadata.
4. Start the UCL 857911 bundles (≈ 7 GB) and the UK HPI CSV.
5. Download ONSPD (latest), IMD 2025 and LSOA 2021 boundaries: the join keys for everything else.
6. Download DESNZ LSOA domestic electricity and gas 2010–2024.
7. Fetch Open-Meteo weather and PVGIS yields per postcode district (cache every response).
8. Download conservation-area and listed-building files from planning.data.gov.uk.
9. Later: London smart meters (10 GB, stream with DuckDB), Electrification of Heat SN 9050 summary, CubiCasa5k, one Octopus time-of-use tariff for tariffs.json.

## The EPC live tables (MHCLG): how they are used

The GOV.UK *Live tables on Energy Performance of Buildings Certificates* are quarterly ODS tables of EPC **lodgements** for England and Wales by local authority × quarter (D1: rating band counts; D3: floor area, modelled energy kWh/m², CO2, fuel cost; EB4: property type; EB-tables for existing dwellings only). They are aggregates, not homes, so they are used to anchor and check the per-home data, not to train per-home models:

| Use | Table | How |
|---|---|---|
| Population calibration targets | D1, EB4 | Rake our EPC sample and the synthetic population so band mix, property-type mix and floor area per local authority match the official totals for the same period. |
| Download / mapper check | D1 | Our certificate counts per LA × quarter × band must reconcile with D1; a gap > 2% flags missing files, opt-outs or mapping errors. |
| Twin validation | D3 | The twin's standard mode (SAP-like occupancy) aggregated per LA vs D3 mean modelled kWh/m² and fuel cost: an official check on the physics. |
| Score peer context | D1, D3 | "Your home vs your area": local-authority band shares and median modelled kWh/m² in ScoreResult.peer_context. |
| Synthetic-data validation | D1, D3, EB4 | Band, type and kWh/m² distributions of the synthetic homes vs the official tables per LA (chi-square, KS), reported in the evaluation. |
| Dashboard layer | D1, D3 | Official LA layer beside our modelled LSOA layer: share A–C vs D–G, trend over quarters, mean modelled kWh/m², CO2 per dwelling. |

Caveats handled: counts are lodgements in a period (repeat certificates, opt-outs), not the dwelling stock; a Q1 2026 local-authority allocation error was revised in May 2026; D3 averages were corrected in December 2023 (recompute averages from totals as the table notes explain).

## How the data joins

| Level | Join | Datasets |
|---|---|---|
| Postcode / LSOA | ONS postcode directory → LSOA 2021, LA code, region | EPC records, UCL sales, weather (by district), planning, off-gas, IMD, DESNZ LSOA |
| Local authority × quarter | LA code | EPC live tables ↔ our EPC sample, synthetic population, twin standard-mode output, dashboard |
| Archetype | type × age band × floor-area band × region | NEED (no postcode) ↔ EPC homes, EBM training, synthetic validation |
| Per home (synthetic) | real EPC home + real weather → twin + behaviour noise | monthly usage + labels |

# 7. Uncertainty: every number is a distribution

Bayesian inference happens in two places: calibrating each home's physics on its bills (Part 1) and the value model on real sales (Part 4). Everything downstream **propagates** those posteriors: each output is computed once per posterior sample, then summarised with contracts.summarise. Intervals are never added together; the samples carry the correlations.

| Part | Fields | Why |
|---|---|---|
| Centre | mean · median · mode | They differ when the result is lopsided |
| Spread | standard deviation · IQR | How uncertain |
| Percentiles + intervals | P1–P99 · 50% and 90% credible intervals | Full shape; the 90% interval is the headline |
| Tails | labelled tail probabilities · mean of the worst 5% · skewness | What could go wrong, and how badly |
| Shape + meta | histogram · unit · n_samples · headline (mean or median) | Density plot; traceability |

| Mean | Median | Mode | Std | P5 | P95 | P(saving > 0) | P(rise > £100) |
|---|---|---|---|---|---|---|---|
| −£35 | −£42 | −£55 | £88 | −£180 | +£110 | 0.38 | 0.23 |

Example (illustrative): heat-pump bill change on SH2. The mean hides that about 1 in 4 homes like this could see the bill rise by over £100/yr while carbon falls with near certainty.

# 8. Key mechanisms

## Improvement ladder: what each retrofit adds

Rules + Jev decide what is possible; the twin decides what it is worth. Step k's effect = twin(steps 1..k) − twin(steps 1..k−1) on every sample, fabric first. The steps add up exactly to the package (one run). Verdicts are probabilistic: Recommended if P(suitable) ≥ 0.8 and P(step adds ≥ 3% of purchased energy or ≥ 2 points) ≥ 0.8. Thresholds to be agreed.

| # | Add (SH2, illustrative) | Gas kWh/yr | Elec kWh/yr | Potential | Score | CO2 t | Bill £ | Verdict |
|---|---|---|---|---|---|---|---|---|
| 1 | Cavity wall insulation | 2,600 | 0 | 17% | 66 | 0.48 | +207 | Recommended |
| 2 | Loft top-up to 270 mm | 900 | 0 | 23% | 68 | 0.16 | +72 | Recommended |
| 3 | Draught proofing | 400 | 0 | 26% | 69 | 0.07 | +32 | Optional |
| 4 | Smart thermostat + TRVs | 500 | 0 | 29% | 71 | 0.09 | +40 | Recommended |
| 5 | Replace old double glazing | 450 | 0 | 32% | 72 | 0.08 | +36 | Not worth adding |
| 6 | Air-source heat pump | 7,150 | −2,300 | 64% | 80 | 0.86 | −35 | Depends on goal |
| 7 | Solar PV 3 kWp | 0 | 1,100 | 71% | 86 | 0.22 | +290 | Recommended |
| | **Package (one run)** | **12,000** | **−1,200** | **71%** | **86** | **1.96** | **+641** | |

## Improvement potential (PS §3)

- 100 × (E_baseline − E_after) / E_baseline on purchased energy (grid import + gas), E_after from all feasible retrofits run together; a distribution with the per-fuel split.
- Needs-survey items reported separately; baseline 0 or missing → unavailable + reason; homes ranked by posterior mean with P(top 10%).

# 9. Evaluation and submission

| Output | Metric (held-out, property splits) | Baseline | Part |
|---|---|---|---|
| Energy prior | MAE/MAPE per fuel, 90% coverage | NEED median, XGBoost | 2 |
| Calibration | coverage ≈ 90%, CRPS, R-hat; standard mode vs EPC D3 by LA | degree-day OLS | 1 |
| Rating after | MAE of SAP change on EPC before/after pairs | median band shift | 2 |
| Recommendations | precision@k vs EPC assessor lists; 0 infeasible | recommend all feasible | 2, 5 |
| Improvement ranking | Spearman / NDCG vs labels | rank by total kWh | 2 |
| Score | 0 monotonicity violations; band-probability calibration | — | 3 |
| Value | coverage, pinball loss; β vs published | OLS without rating | 4 |
| GNN area model | MAE on held-out LSOAs (spatial blocks) | EBM without graph | 0/dashboard |
| Photo capture | room-area error, tag accuracy, % fields corrected | manual tagging | 1 |
| LLM | groundedness ≥ 90%, accuracy ≥ 90%, unsupported < 5% | no validator | 5 |
| Synthetic data | KS vs NEED; band/type mix vs EPC live tables | — | 0 |

| PS | Requirement | Part |
|---|---|---|
| §1 | Core rule; what-if reruns models first | 5 + orchestrator |
| §2 | Score with fixed weights, documented monotonic sub-scores, contributions | 3 |
| §3 | Poor homes, key issues, improvement potential per fuel | 3 + 2 |
| §4 | Feasible, prioritised retrofits with suitability, saving, reason | 5 + 2 |
| §5 · §6 | Per-fuel savings with interactions; carbon from documented factors | 1 + 2 |
| §7 | Indicative value range, before vs after | 4 (+ 2 rating) |
| §9 | Data, splits, leakage, documented synthetic labels | 0 |
| §10 | Customer memory changes recommendations | 5 |
| §13 | A prototype · B code · C models · D score doc · E LLM doc · F this document · G eval · H RAI · I test predictions | all |

# 10. Build plan

## Work breakdown

| Track | Sequence | Starts on |
|---|---|---|
| **Coupled core** (Parts 1 + 2, one team) | physics twin (JAX) → rating estimator → synthetic generator → Bayesian calibration → ladder, potential, Pareto → electricity layer | RdSAP defaults, SH1–SH3 examples |
| Part 0 Data | EPC access + downloads → mapper → splits → DuckDB → live tables + DESNZ + smart meters | day 1 |
| Part 3 Score | score + monotonic tests → key issues → peer context → projected score | recorded inputs |
| Part 4 Value | UCL sales prep → Bayesian hedonic → before/after → green twins | UCL sample; stub rating |
| Part 5 Feasibility & LLM | rules engine → mock server → chat + validator → eval set → Jev → memory | mocks |
| Experience | app shell on mocks → 3D twin v1 → capture editor → dashboard | mocks, generated types |

| Checkpoint | Done when |
|---|---|
| 1. End-to-end skeleton | SH1–SH3 flow through the real twin → score → retrofits → API → UI |
| 2. Complete submission | Bayesian distributions, rating estimator, value, grounded LLM, evaluation, docs |
| 3. Demo build (freeze) | 3D twin + capture, Pareto + loan, Lloyds dashboard, memory |

## Working rules (hackathon: speed first)

- Speed beats polish: thinnest end-to-end path first, mocks from day one, reuse and hardcode for the demo, timebox at ~45 minutes, commit small and often.
- Tests only where judges look: contracts, monotonicity, per-fuel signs, null potential at baseline 0, the LLM validator, no mock in the demo, types.ts in sync.
- Non-negotiables: contracts in and out; provenance; Distributions via summarise; missing ≠ 0; rerun never sum; the LLM never makes a number; value always indicative; no keys in git.
- One UI look: shared theme and components (NumberWithRange, DistributionCard, ScoreWaterfall), brand green #0b6e4f / blue #2a6fdb, colour-blind-safe scales, every number with unit + range.

## Open decisions

| Decision | Options | Recommendation |
|---|---|---|
| Twin ambition (library: JAX) | A: monthly steady-state first · B: differentiable RC graph (diffrax) first | A for checkpoint 1; upgrade behind simulate() |
| Verdict thresholds | gain threshold, probability cut-offs, cost per kWh | Agree once; record in SCORE doc |
| Default goal | bills or carbon | Bills by default; carbon toggle |
| Time-of-use tariff source | which published tariff to cite | Pick one public tariff; until then null |